



Master HVAC Engineering

The Complete Professional Guide

Refrigeration • AC Systems • Chillers • Installation • Safety

Awet G. Nway

Robotics, Automation & MEP Engineering Specialist

www.ayetechub.com

AYE Tech Hub • Engineering Education & Services

Welcome to the Complete HVAC Guide

Heating, Ventilation & Air-Conditioning (HVAC) is the engineering discipline behind thermal comfort, indoor air quality, and industrial process cooling. This guide is a structured curriculum that takes you from the underlying thermodynamics of the refrigeration cycle to the installation, maintenance and safety practices used by professional technicians on residential, commercial and industrial systems.

Each module is self-contained, illustrated with schematic diagrams, and includes a practical checklist you can apply on the job. Use the guide as a study companion or as a field reference.

What you will learn

Module	Topic	Outcome
1	Core Refrigeration Cycle	Explain heat transfer and the 4-stage vapor-compression loop
2	Split & Window AC Systems	Differentiate residential AC types and their components
3	Industrial Chiller Systems	Select air-cooled vs water-cooled for large loads
4	Installation & Maintenance	Execute a safe, code-compliant split-AC install
5	Safety & Technical Tools	Apply PPE and use the standard HVAC tool kit

i

INFO How to use this guide

Work through the modules in order. Pay close attention to the diagrams — they encode the physical relationships you must visualise on the job. Complete the self-check quiz at the end before applying the techniques on a live system.

1

Master the Core Refrigeration Cycle

Every HVAC system — from a 1.0-ton bedroom split unit to a 2,000-ton centrifugal chiller — is built around the same physical principle: the vapor-compression refrigeration cycle. A working fluid called a refrigerant continuously circulates between two heat exchangers, absorbing heat where you want it cold and rejecting it where you can afford to dump it.

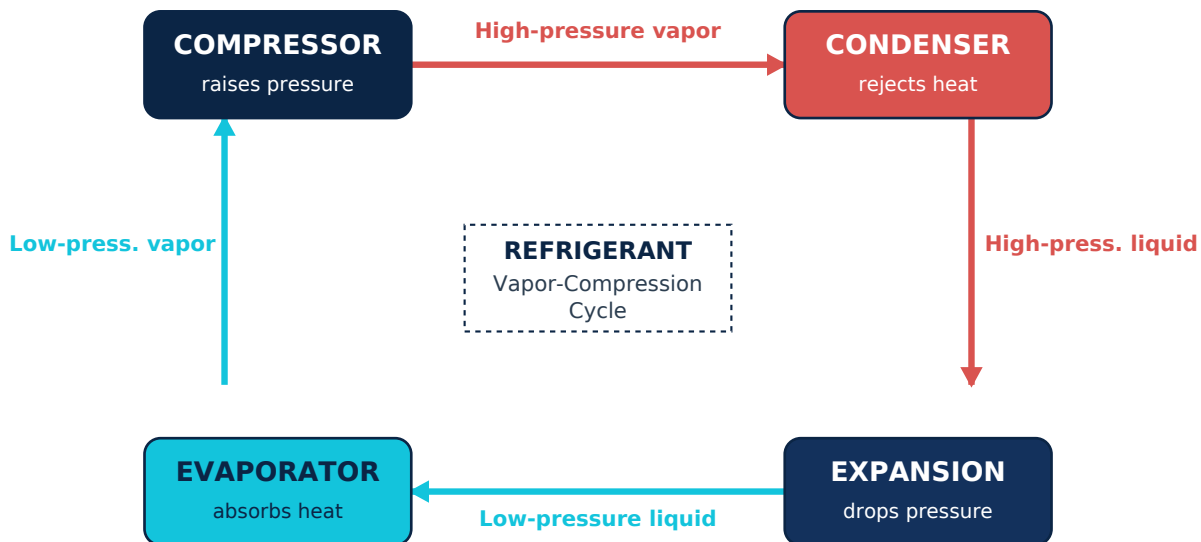


Figure 1.1 — The four-component vapor-compression cycle

The four essential components

Component	Function	Refrigerant state at outlet
Compressor	Raises pressure & temperature of vapor; drives the cycle	High-pressure, superheated vapor
Condenser	Rejects heat to ambient air or cooling water; vapor condenses	High-pressure, sub-cooled liquid
Expansion valve	Drops pressure abruptly; refrigerant flashes and cools	Low-pressure, two-phase mixture
Evaporator	Absorbs heat from the conditioned space; liquid boils	Low-pressure, slightly superheated vapor

✓ TIP Engineering rule of thumb

Coefficient of Performance, $COP = Q_{cooling} / W_{compressor}$. A well-tuned modern split AC achieves $COP \approx 3.2-4.0$ in cooling mode. If a unit's measured COP drops below ~ 2.5 , suspect refrigerant charge, dirty coils, or compressor wear.

2 Split and Window AC Systems

Residential and light-commercial cooling is dominated by two configurations: the split system, which separates the noisy compressor and condenser into an outdoor unit, and the window unit, which packages everything into a single chassis that sits in a window or wall opening. Both implement the same refrigeration cycle from Module 1, only their physical layout differs.

Split AC — two-piece architecture

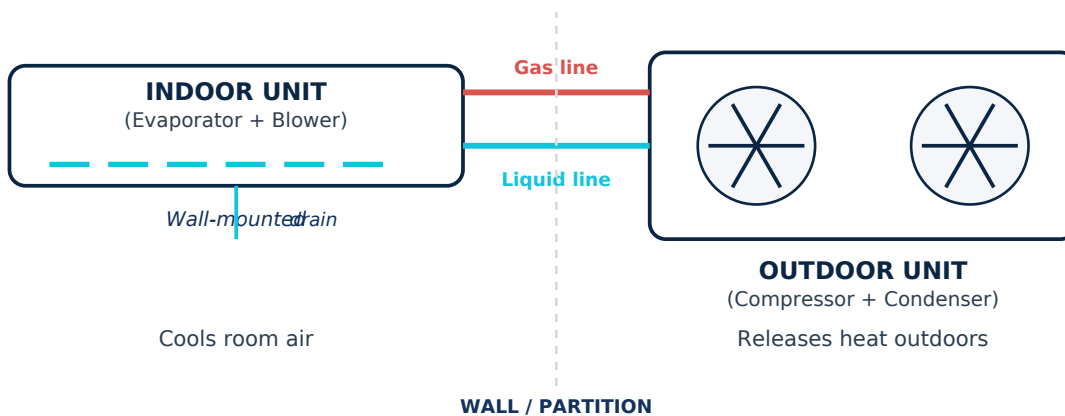


Figure 2.1 — Split AC: indoor evaporator, outdoor condenser, linked by insulated copper

Window AC — single-chassis design

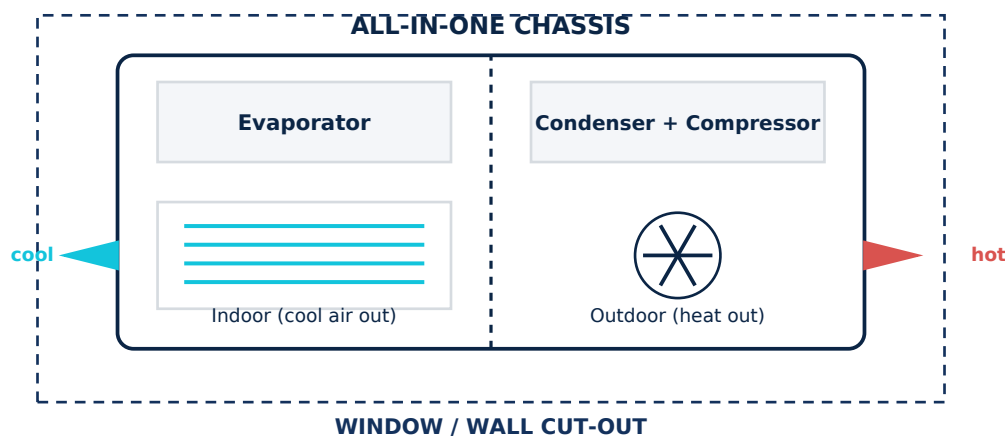


Figure 2.2 — Window AC: one cabinet, divided internally between cold and hot sides

2 Split and Window — Detailed Comparison

Attribute	Split AC	Window AC
Typical capacity	0.75 – 5 TR	0.5 – 2.5 TR
Installation	Indoor unit + outdoor unit, connecting copper pipes & cable, vacuum & charge	Single chassis slid into window/wall sleeve
Noise level	Low indoors (compressor is outdoors)	Higher — compressor inside the room
Aesthetics	Discreet wall unit	Visible from inside and outside
Efficiency (SEER)	16 – 26 (inverter models)	10 – 14
Installation cost	Higher — needs piping & vacuum work	Lower — minimal mounting
Best for	Bedrooms, offices, multi-room (multi-split)	Single rooms, rental units, retrofits
Maintenance	Filter, indoor coil, outdoor coil, gas top-up	Filter, single coil, drainage

Maintenance schedule (residential)

Task	Interval
Wash air filter	Every 2-4 weeks
Clean indoor evaporator coil	Every 6 months
Clean outdoor condenser coil & fins	Every 6-12 months
Check drain line / pan	Every 3 months
Measure superheat & sub-cooling	Annually (or on complaint)
Inspect electrical contactors & capacitors	Annually

! WARNING Refrigerant handling

Never vent refrigerant to atmosphere. In most jurisdictions a refrigerant-handler certificate is mandatory before opening a sealed system. Always recover refrigerant into an approved recovery cylinder before brazing or component replacement.

3

Industrial Chiller Systems

When the cooling load exceeds the practical limit of unitary AC (≈ 25 TR per system), the industry moves to chillers: large machines that cool water (or a brine), which is then pumped through air-handling units, fan-coils, or process equipment. Chillers are classified by how they reject heat from the condenser side.

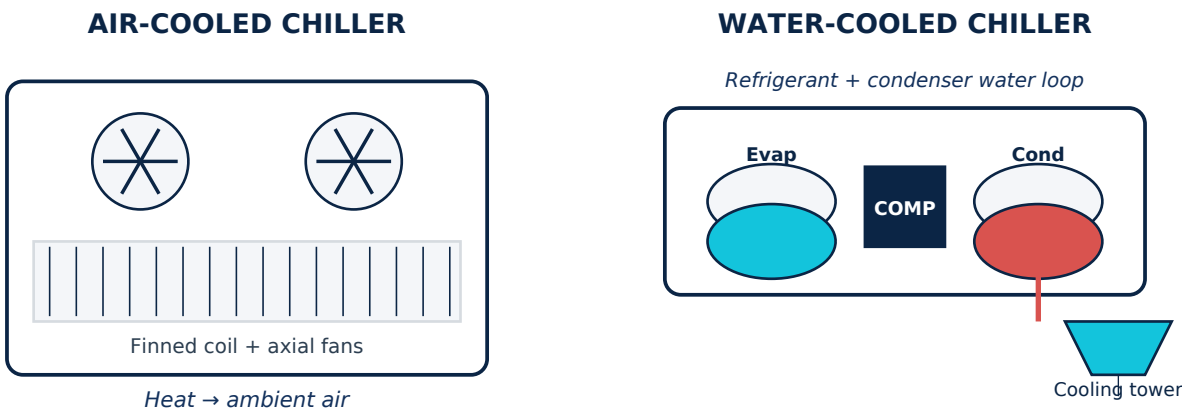


Figure 3.1 — Air-cooled (left) vs water-cooled chiller (right)

Air-cooled vs water-cooled — selection criteria

Criterion	Air-cooled chiller	Water-cooled chiller
Heat rejection	Ambient air via finned coil + axial fans	Condenser water $\rightarrow$ cooling tower
Typical capacity	20 – 500 TR	100 – 3000+ TR
Efficiency (kW/TR)	1.0 – 1.4	0.5 – 0.7 (significantly better)
Footprint	Single packaged unit, rooftop friendly	Chiller + cooling tower + pumps + piping
Water consumption	None	Significant make-up & blowdown
First cost	Lower	Higher (more components)
Best application	Small/medium loads, water-scarce sites	Large commercial / industrial, year-round operation

i INFO Compressor types inside chillers

Scroll (up to ~ 60 TR) — quiet, reliable, modular. Screw (60 – 500 TR) — workhorse for commercial buildings. Centrifugal (300 – 3000+ TR) — highest efficiency, used in large district cooling plants.

4

Professional Installation & Maintenance

A correct installation is more than 50% of a long, trouble-free service life. The most common failures of new split ACs in the field are traceable to three preventable errors: insufficient evacuation, leaks at flare connections, and incorrect refrigerant charge. The procedure below is the professional sequence used on R-32 / R-410A split systems.

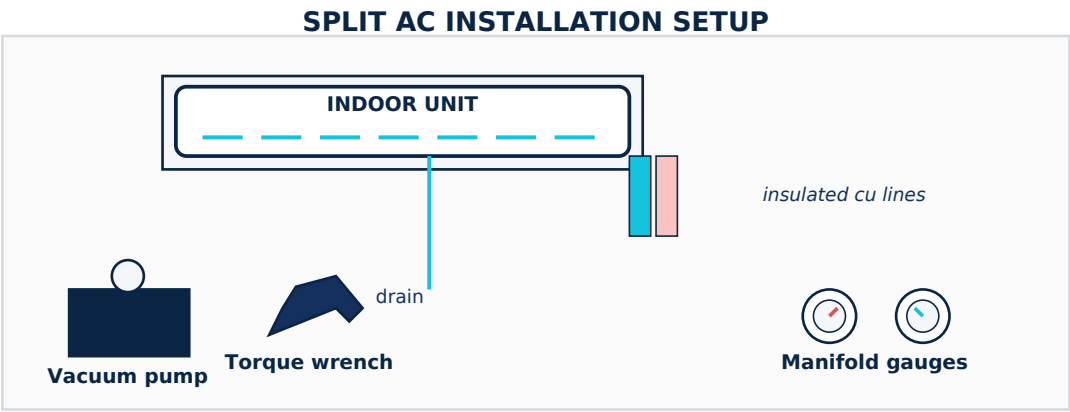


Figure 4.1 — Split AC installation setup with manifold gauges and vacuum pump

Step-by-step installation sequence

1. Site survey	Confirm wall structure, electrical supply (voltage, breaker size), drainage route, and clearances around the outdoor unit (≥ 300 mm sides, ≥ 600 mm front).
2. Mount indoor unit	Install the wall plate using a spirit level. Mark and drill the wall pass-through with a slight downward slope toward the outside for drainage.
3. Mount outdoor unit	Use vibration-isolation mounts on a level pad or wall brackets rated for the unit weight + 30%.
4. Run pipework	Route the insulated copper liquid and gas lines, drain hose and interconnect cable through the sleeve. Flare both ends with a proper flaring tool — no nicks, no oil contamination.
5. Pressure & leak test	Pressurise with dry nitrogen to ~ 32 bar (R-410A) or per manufacturer. Hold and check with soap solution or electronic leak detector. No drop in 15 minutes = pass.
6. Evacuate	Pull a vacuum to below 500 microns (≈ 0.67 mbar). Hold for at least 15 minutes — pressure must not rise. A failed vacuum hold indicates a leak or trapped moisture.
7. Release factory charge	Open both service valves fully with the correct hex key. Confirm low/high pressures and superheat on the manifold.
8. Commission	Run the unit for 15 minutes. Measure supply-air temperature drop (typically 8 – 12 °C below return-air). Verify drain flow and check for vibration.

4 Troubleshooting Reference

Use this matrix as a first-pass diagnostic when a system arrives with a complaint. Always confirm the symptom with measurements before changing parts.

Symptom	Likely cause	First check
Insufficient cooling	Low refrigerant charge / leak	Measure superheat & sub-cooling; soap-test joints
Indoor unit icing up	Low airflow or low charge	Filter, blower speed, evaporator coil cleanliness
Outdoor unit not running	Capacitor / contactor failure	Test run-capacitor μF value; listen for contactor click
Compressor short-cycles	Dirty condenser, low voltage, high pressure switch	Clean coil; measure supply voltage under load
Water leaking from indoor unit	Blocked drain line or wrong slope	Flush drain; verify $\geq 1\%$ downward slope
Loud humming, no start	Seized compressor or failed start cap.	Test winding resistance; replace start capacitor
Burning smell	Electrical short / overheated motor	Isolate power immediately — inspect wiring

× **DANGER** Electrical & burn hazards

Compressor discharge lines exceed 90 °C in normal operation. Capacitors retain dangerous charge for minutes after disconnection — always short-discharge through a resistor before handling. Never work on live circuits; lock-out / tag-out the breaker.

5

Safety & Technical Tools

Professional HVAC work involves four overlapping hazards: electrical (240–415 V mains and stored capacitor energy), mechanical (rotating fans, working at height, lifting heavy outdoor units), thermal (hot discharge lines, brazing), and chemical (refrigerants and brazing fumes). Every site visit begins with PPE and a pre-task risk check.

ESSENTIAL HVAC PPE & TEST EQUIPMENT



Figure 5.1 — Standard PPE and field test equipment

Personal protective equipment (PPE)

PPE Item	When required
ANSI Z89 hard hat	On any installation site or under overhead work
Safety goggles / face shield	Brazing, drilling, refrigerant handling
Insulated work gloves (Class 0)	Any live-circuit verification
Cut-resistant gloves	Sheet-metal duct work, copper handling
Steel-toed safety boots	All field work
Half-mask respirator	Brazing in confined spaces, coil cleaning
Hearing protection	Within 3 m of operating chillers / cooling towers

5 Standard HVAC Tool Kit

Tool	Purpose
Manifold gauge set (R-32 / R-410A)	Read suction / discharge pressures; charging
Vacuum pump (≥ 2 CFM, 2-stage)	Evacuate moisture and non-condensables
Digital micron gauge	Verify deep vacuum below 500 microns
Electronic leak detector / soap solution	Locate refrigerant leaks at joints
Flaring & swaging tool kit	Prepare copper tube ends for service connections
Tube cutter & deburring tool	Clean cuts on copper without burrs
Digital multimeter (True-RMS)	AC voltage, current, capacitance, continuity
Clamp meter (AC/DC, 600 V)	In-line current measurement during operation
Torque wrenches (15 - 70 N·m range)	Tighten flare nuts to manufacturer spec
Digital thermometer / thermocouple set	Measure superheat, sub-cooling, ΔT across coil
Refrigerant recovery machine + cylinder	Mandatory before opening a sealed system
Hex / Allen key set (metric)	Open service-valve cores

✓ TIP Calibration & care

Calibrate digital gauges and micron meters annually. Drain and refill vacuum-pump oil after every job — contaminated oil destroys the pump's ability to pull a deep vacuum, which is the single biggest cause of premature compressor failure.



Self-Check Quiz

Test your understanding. Cover the answer key below the line and check yourself.

Q1. Name the four core components of the vapor-compression refrigeration cycle, in order of refrigerant flow starting from the compressor outlet.

Q2. A split AC drops only 4 °C across the indoor coil. Give three plausible causes you would check first.

Q3. Why is a deep vacuum (< 500 microns) essential before charging a new split system?

Q4. For a 1,200 TR commercial building load, would you specify air-cooled or water-cooled chillers? Justify your choice.

Q5. List the minimum PPE required when brazing a copper joint indoors.

Answer key

A1. Compressor → Condenser → Expansion Valve → Evaporator → (back to compressor).

A2. Dirty filter or coil restricting airflow; low refrigerant charge; blower speed set too low / fan motor weak; iced evaporator from previous low-charge operation.

A3. Moisture inside the system reacts with refrigerant and lubricant to form acids that attack compressor windings; trapped non-condensables raise head pressure and reduce capacity.

A4. Water-cooled. At 1,200 TR the lifetime energy savings of a 0.5–0.7 kW/TR water-cooled plant vs a 1.0–1.4 kW/TR air-cooled bank far outweigh the extra first cost of cooling towers and pumps.

A5. Safety goggles or face shield, flame-resistant gloves, long sleeves, steel-toed boots, and a half-mask respirator. A fire-watch and CO₂ extinguisher must be on hand.

i INFO Continue your learning

Visit www.ayetechub.com for the full HVAC course series, downloadable design calculators (cooling load, duct sizing, pipe sizing), and step-by-step video walk-throughs. AYE Tech Hub also delivers HVAC, MEP and Solar engineering services across Tigray, Ethiopia.



Stay Connected With AYE Tech Hub

About the author

Awet G. Nway is a Robotics, Automation and MEP Engineering specialist with hands-on experience in industrial automation, machine foundation supervision, production-and-maintenance team leadership, and HVAC systems. He is the founder of AYE Tech Hub, an engineering education and services platform serving learners and businesses across Ethiopia and beyond.

Contact & resources

Website	www.ayetechub.com
Email	awetgknway@gmail.com
Social handles	@ayetechub @ayetechpro
Platforms	YouTube • Telegram • Instagram • LinkedIn • Threads • TikTok • Facebook
Services	Engineering education • PLC & Automation • MEP • HVAC • Solar • CAD/Revit • Interior & Architectural Design
Languages	English • Amharic • Tigrinya

Engineer the future. Learn it once. Apply it anywhere.

— AYE Tech Hub | Empowering African Engineering —